Fundamentals of Probability

Statistics = Descriptive + Inferential : from sample to population

Probability →

Key distinction: DATA : what "happened", we observe
vs
POSSIBILITIES : what could happen, unobserved

the DATA GENERATING PROCESS (DGP)

↳ underlying (unobserved) system that generates the data

↳ we want to learn/discover the DGP and to model/represent it

- Random Variables
- Prob. Dist.
- Relationships
- Noise

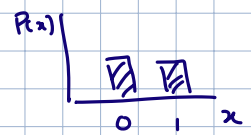
Random Variable: X → R.V.
x → specific value

random process → outcomes → numbers

Prob. Distribution: table, graph, function representing all possible values of X and how likely they are

Discrete RV $P(X=x) = P(x)$

x	$P(x)$
0	0.5
1	0.5

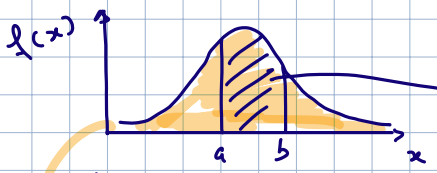


$P(x) = 0.5$

$0 \leq P(x) \leq 1$

$\sum P(x) = 1$

Continuous RV $f(x)$



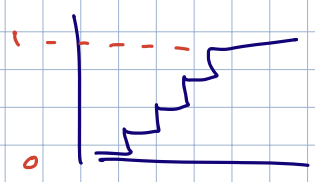
$P(X = a) = 0$

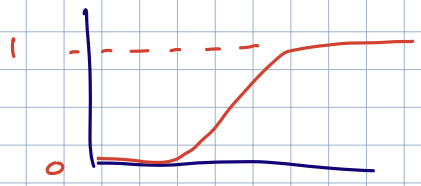
$P(a \leq X \leq b) = \int_a^b f(x)dx$

$f(x) \geq 0$

$\int_{D_x} f(x)dx = 1$

Cumulative Dist. CDF $F(x) = P(X \leq x)$





Joint Distributions : $P(X=x, Y=y) = P(x,y)$ $\quad f(x,y)$

AND

x \ y	1	2	3	
1	0	0.15	0.3	0.45
2	0	0	0.25	0.25
3	0.1	0.15	0.05	0.3
	0.1	0.3	0.6	

marginal dist. of Y

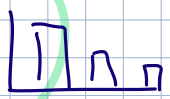
x	P(x)
1	0.45
2	0.25
3	0.3

y	P(y)
1	0.1
2	0.3
3	0.6

Conditional Dist. $\quad P(X=x \mid Y=y) = \dfrac{P(x,y)}{P(y)}$

$$P(X=1 \mid Y=3) = \frac{P(X=1, Y=3)}{P(Y=3)} = \frac{0.3}{0.6} = 0.5$$

x	P(x | Y=3)
1	0.3/0.6
2	0.25/0.6
3	0.05/0.6



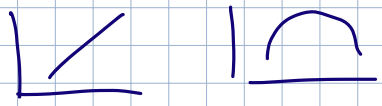
x	P(x | Y=1)

x	P(x | Y=2)

Independence : $\quad P(X=x \mid Y=y) = P(x)$

for all x and y

Dep.



3 - Fundamentals of Probability

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Agenda

Today we'll build the foundation for statistical thinking by exploring:

1. Random variables: how we represent uncertain outcomes
2. Probability distributions: how we model uncertainty
3. Conditional probability: how we reason about outcomes given other information
4. Joint and marginal relationships: how variables interact
5. Common distributions used in business analytics

Recommended readings: Wooldridge Appendix B

What *Is* Probability?

> *"Probability is the branch of mathematics concerning events and numerical descriptions of how likely they are to occur."* *(from Wikipedia)*

In business analytics, we rarely have complete information.

- We make decisions with samples, not populations.
- We test strategies based on incomplete and noisy data.

Probability gives us the **tools to deal with uncertainty**.

Why Review Probability?

Business analytics often involves **making decisions under uncertainty**:

- Will a customer make a purchase?
- How many units will we sell next week?
- Will a new strategy improve conversion rates?

Probability theory gives us the tools to:

- Represent uncertain outcomes, e.g., Purchase = Yes or No
- Quantify how likely they are, e.g., P(Purchase = Yes) = 0.65
- Compare risks and rewards, e.g., expected revenue vs. cost

It helps us separate two key ideas:

- What **could happen** – the *probability model*
- What **did happen** – the *observed data*

What About Simulation and Modern Tools?

Many modern analytics methods rely on **simulation** instead of explicit mathematical models and formulas:

- Bootstrapping
- Permutation tests
- Machine learning algorithms

(You'll learn all of these in later courses this year.)

But even these rely on **key probability concepts**:

- Sampling distributions (bootstrapping)
- Random assignment under the null (permutation tests)
- Expectations, independence, and uncertainty (ML performance)

You still need probability to understand how modern methods work and when to trust their results.

Why We Start With Probability

Probability is the foundation for understanding uncertainty, testing hypotheses, and interpreting variation in data.

It gives you the tools to:

- Understand the **assumptions** behind statistical models
  - e.g., independence, normality, constant variance
- Evaluate the **reliability of statistical inferences**
  - e.g., Is this effect real or just random variation?
- Make sound **decisions under uncertainty**
  - e.g., How likely is this outcome? What's the risk of being wrong?

Later, you'll learn tools that are more flexible and assumption-light but, under the hood, they still rely on the **logic of probability**.

Data collection



1. Random variables and their probability distributions

The Survey Data (from 2024/25/26)

```r
# Load packages
library(tidyverse)

# Load data
broccoli <- read_csv("clean_broccoli_survey_data.csv")
```

```r
# Glimpse data
glimpse(broccoli)
```

```
Rows: 109
Columns: 3
$ LikeBroccoli <dbl> 3, 7, 1, 10, 10, 8, 7, 7, 3, 3, 10, 7, 8, 7, 8, 7, 10, 7,~
$ WillingToPay <dbl> 4.34, 3.81, 0.00, 1.01, 2.60, 4.16, 1.50, 1.45, 0.05, 0.8~
$ WillingToBuy <dbl> 1, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 0, 0, ~
```

```r
# Display the first few rows of the dataset
head(broccoli, 3)
```

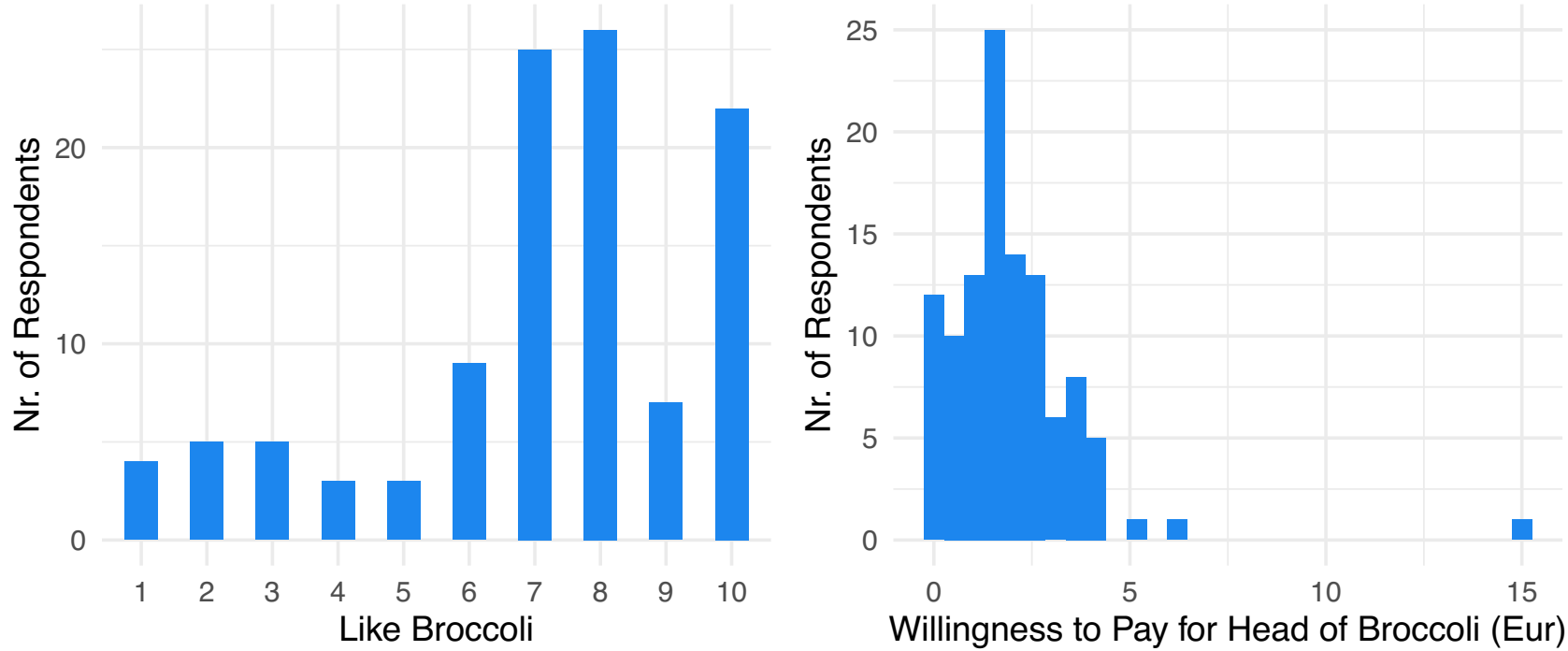
```
# A tibble: 3 x 3
  LikeBroccoli WillingToPay WillingToBuy
         <dbl>        <dbl>        <dbl>
1            3         4.34            1
2            7         3.81            1
3            1         0               0
```

Frequency distributions

```
# Frequency distribution - Like
p1<- ggplot(data = broccoli, aes(x = factor(LikeBroccoli))) +
  geom_bar(fill = "dodgerblue2", width = 0.5) + theme_minimal() +
  labs(x = "Like Broccoli", y = "Nr. of Respondents")

# Frequency distribution - WTP
p2 <- ggplot(data = broccoli, aes(x = WillingToPay)) +
  geom_histogram(fill = "dodgerblue2") + theme_minimal() +
  labs(x = "Willingness to Pay for Head of Broccoli (Eur)", y = "Nr. of Respondents")

grid.arrange(p1, p2, ncol = 2)
```



From Data to Theory

We've seen how students **vary** in their broccoli preferences and willingness to pay.

Now let's shift our thinking:

- Instead of just **looking at data**, can we describe a **process** that might have generated it?
- Can we imagine this process includes some **unpredictable elements**?

We are moving from **describing what we saw** to **explaining how it might have come about**.

Introducing the Data Generating Process (DGP)

The **Data Generating Process** (DGP) is a theoretical model for how data comes into existence.

It represents the **underlying system** that generates the values we observe and includes:

- **Random variable(s)**: numeric values used to represent uncertain outcomes.
- **Probability distribution(s)**: a model that tells us how likely each possible value of a variable is.
- **Structural relationships**: how variables relate to one another.
- **Noise**: unexplained variation or randomness in the process.

In real life, the DGP is usually **unknown** but by making assumptions, we can **build models** that approximate it.

What Is a Random Variable?

A **random variable** assigns a number to the outcome of a process involving uncertainty.

Two main types:

- **Discrete**: takes specific values (e.g., 1, 2, 3...)
- **Continuous**: can take any value in an interval (e.g., €2.13)

In our survey:

- X = Liking score $\rightarrow$ **ordinal**, treated as **discrete**
- Y = Willingness to pay $\rightarrow$ **continuous**

X and Y aren't spreadsheet columns, they represent **uncertain quantities** that we can model using probability.

Is "Like Broccoli" Really Discrete?

Technically: it's an **ordinal** variable (1–10 scale)

- Has order, but not guaranteed equal spacing

Practically: we **treat it as discrete**

- Allows us to apply probability models
- Just interpret results cautiously

In future surveys, we might ask:

- "How many times did you eat broccoli last week?"

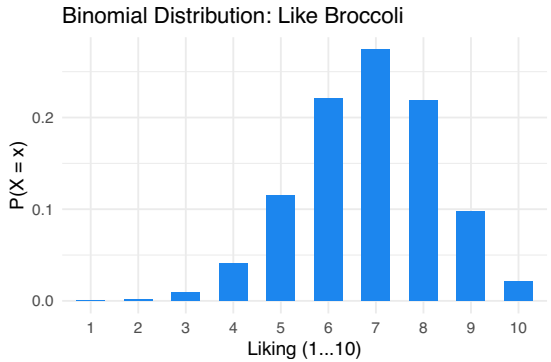
That would give us a **true count variable** — naturally discrete.

What Is a Probability Distribution?

A **probability distribution** tells us how likely each outcome of a random variable is.

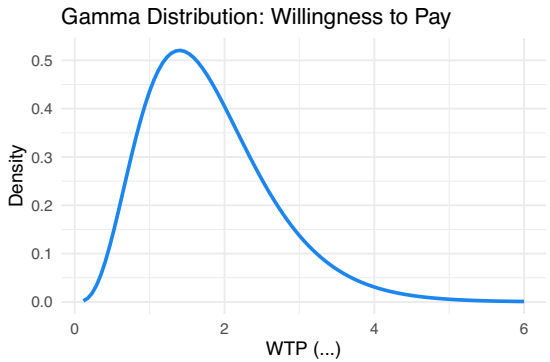
- **Discrete**: Probabilities assigned to each value
  - e.g., P(Liking $= 7$) $= 0.274$
- **Continuous**: Probabilities assigned across a range of values
  - e.g., P($2.00 <$ WTP < 3.00) $= 0.26$

Discrete Case: $P(X = x)$



$0 \leq P(x) \leq 1$ and $\sum_x P(x) = 1$

Continuous Case: $f(x)$



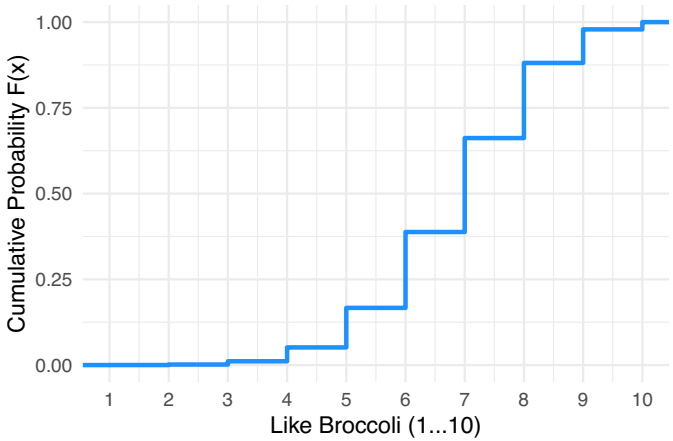
$f(x) \geq 0$ and $\int f(x)\,dx = 1$

Cumulative Distribution Function (CDF)

The **CDF** shows the probability that a random variable is less than or equal to a value:

$$F(x) = P(X \leq x)$$

Discrete example



Continuous example



Use the CDF to answer questions like:

- "What's the chance WTP $\leq$ 1 euro?"
- "What percent of students like broccoli $\geq$ 4/10?"

2. Joint distributions, conditional distributions, and independence

Why Think About Two Variables at Once?

Until now, we've described **one random variable at a time**.
But in data analysis, we often care about how **two variables relate**.

Think of examples like:

- Does liking broccoli relate to willingness to pay?
- Does study time affect exam score?
- Does price affect purchase behavior?

To answer questions like these, we need new tools:

- **Joint distributions**
- **Conditional distributions**
- **Independence**

Joint Distributions (Discrete)

A **joint distribution** shows the probability of different combinations of values for **two random variables**.

For example: *What's the probability that someone likes broccoli = 8 AND would buy it (wtp > market price)?*

This is written:

$$P(X = 8, Y = 1)$$

We'll explore this using **simulated population data** (not something we usually have in real life)!

Table 1: Joint Probability Distribution: LikeBroccoli and WillingToBuy

WillingTo-Buy	1	2	3	4	5	6	7	8	9	10
No (Y=0)	0.1	0.1	0.09	0.05	0.02	0.0	0.0	0.0	0.0	0.0
Yes (Y=1)	0.0	0.0	0.01	0.05	0.08	0.1	0.1	0.1	0.1	0.1

Conditional Distributions

A **conditional distribution** shows how the distribution of one variable changes **given that** we know something about another.

> *Example: Among people who would buy broccoli, how much do they like it?*

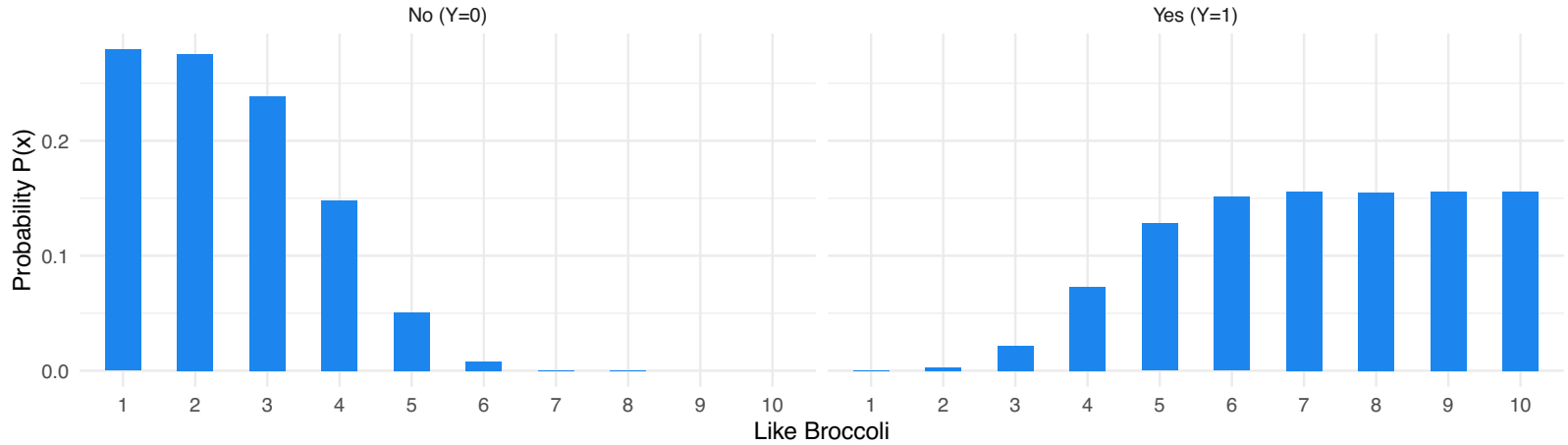
This is a **conditional probability**:

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} \quad \text{for } P(Y = y) > 0$$

It tells us how **liking varies** depending on **buying behavior**, a key part of understanding relationships between variables.

Exploring Conditional Distributions

Let's visualize how the **distribution of liking** changes depending on **willingness to buy** (synthetic pop. data).



We see:

- People who would buy broccoli tend to like it more
- People who wouldn't buy tend to have lower liking scores
- This is evidence of a statistical relationship between the two variables.

What Is Independence?

Two variables are **independent** if knowing the value of one tells you **nothing new** about the other.

This means:

$$P(X = x \mid Y = y) = P(X = x) \text{ for all } x, y$$

In other words, learning something about Y doesn't change the probability of X.

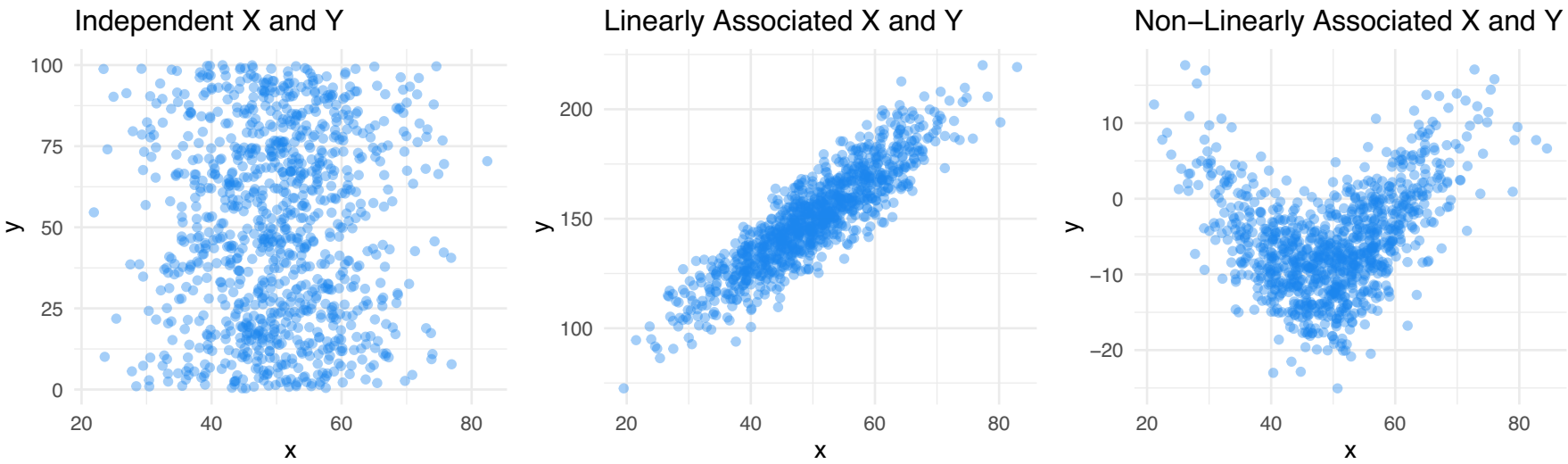
If this doesn't hold, the variables are **dependent**.

This matters because:

- Independence is a common **model assumption**
- Violating it affects how we compute probabilities and make inferences

Independence vs. Dependence

Let's compare different relationships:



3. Features of probability distributions

Marginal Distribution — Center
\ Spread

Joint Distributions — Associations

Why Summarize Distributions?

Probability distributions can be complex. But often, we just want to know:

- What's **typical**?
- How much do values **vary**?
- How are **two variables related**?

Most of the time, we focus on just a few important properties:

- **Center** – where is it centered?
  - Mean, median
- **Spread** – how much does it vary?
  - Variance, standard deviation
- **Association** – how do two variables move together?
  - Covariance, correlation

$E[X]$ mean

mean — R.V.

Population (true) mean of X

Discrete

x	P(x)
0	0.5
1	0.5

Continuous

$$E[X] = \int x f(x)\, dx$$

$0 \times 0.5 + 1 \times 0.5$

$$E[X] = \sum x P(x)$$

$$\sigma^2 = Var = E[(X - \mu)^2]$$

Pop. (true) variance

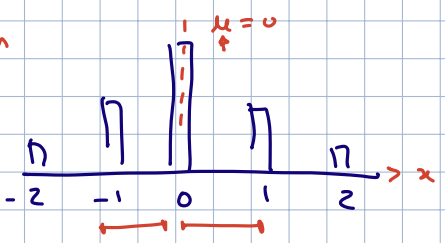
Mean Average

Pop. mean

$$V(X) = \sum (x - \mu)^2 P(x)$$

$$SD = \sigma$$

Distance



( )²

Expected Value: Long-Run Average

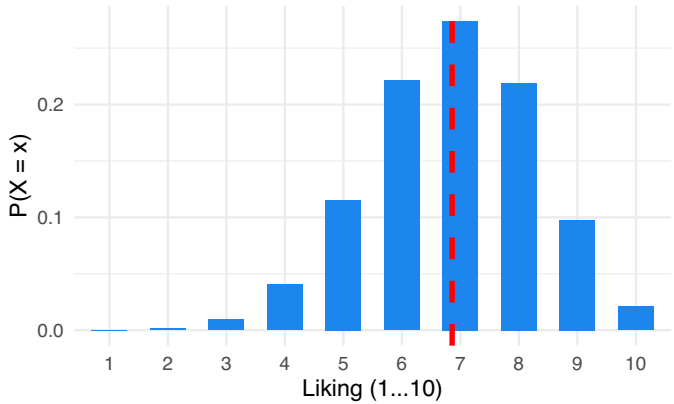
The **expected value** is the long-run average value of a random variable.

Think of it as: if we repeated this process **many times**, what would the **average outcome** be?

$$\mu = E[X] = \begin{cases} \sum_x xP(x), & \text{(discrete)} \\ \int xf(x)\,dx, & \text{(continuous)} \end{cases}$$

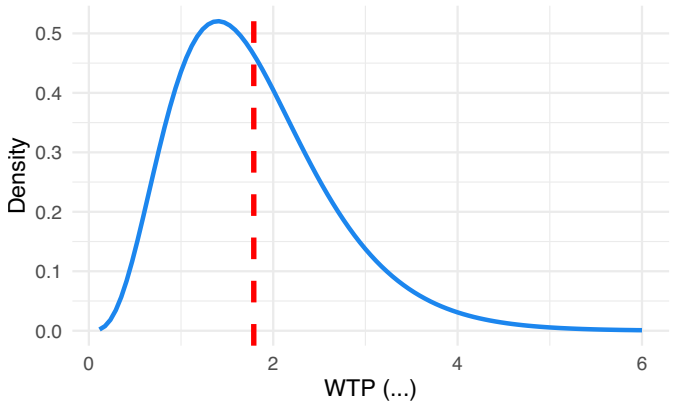
Discrete Case

Binomial Distribution: Like Broccoli



Continuous Case

Gamma Distribution: Willingness to Pay



What Is Variance?

Variance helps answer: on average, **how far do values fall from the mean**?

- Large variance $\rightarrow$ more spread (risky, unpredictable)
- Small variance $\rightarrow$ tightly clustered (predictable, stable)

$$Var(X) = E[(X - \mu)^2] = \sum_x (x - \mu)^2 \, P(x)$$

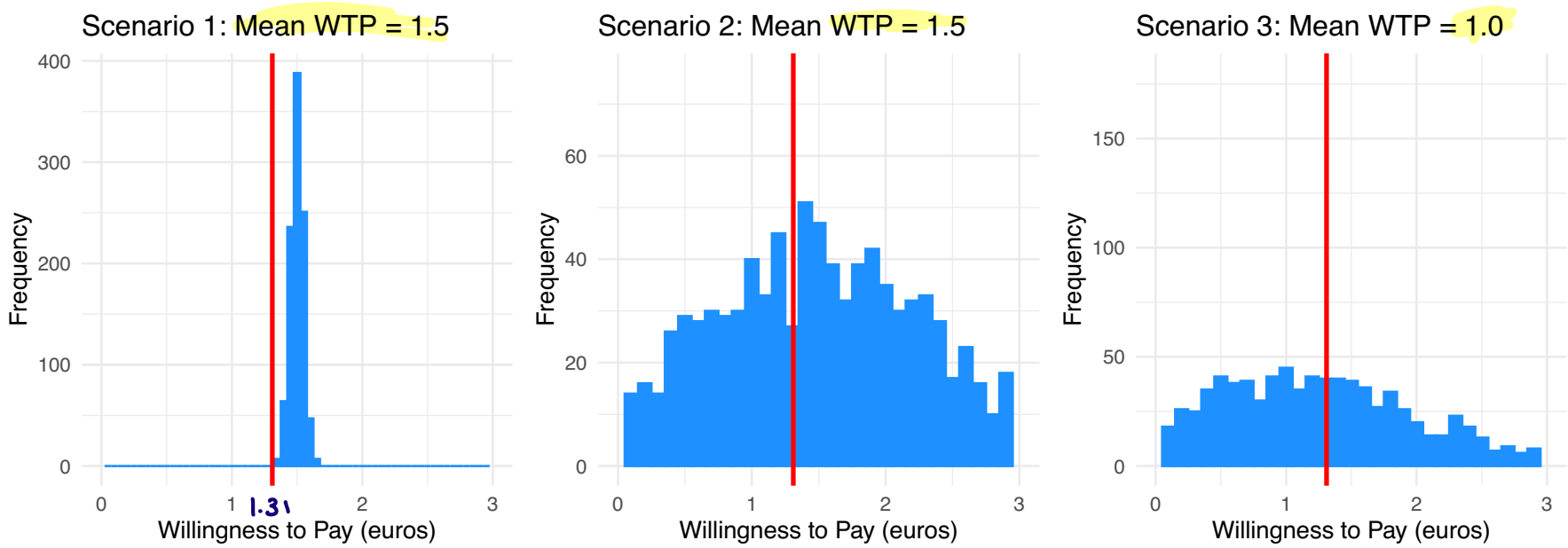
The **variance** of a random variable X, denoted $Var(X)$, σ^2, or σ_X^2, is the **expected value of the squared deviation** from the mean.

The **standard deviation** (SD), denoted σ, is simply the square root of variance - it's in the **same units** as the original variable.

When Does Variance Matter?

You're a broccoli farmer charging **€1.31** per head. You care not just about average **willingness to pay (WTP)** but **how spread out** those values are.

Let's simulate 3 situations:



Why Standardize?

Different variables have different scales, e.g.: WTP is measured in euros, while liking is on a 1–10 scale.

We use **standardization** to put them on the same footing:

$$Z = \frac{X - \mu}{\sigma}$$

Now we can compare apples to oranges using **z-scores**!

The **z-score** of a value tells us how many standard deviations it is from the mean.

It is useful when:

- Comparing variables with different units
- Spotting outliers
- Building models with variables on different scales

Example: Standardizing Willingness to Pay

↳ Sample

```r
# Compute mean and sd of WTP
mean.WTP <- mean(broccoli$WillingToPay)
sd.WTP <- sd(broccoli$WillingToPay)

# Create standardized variable
broccoli <- broccoli |> mutate(z_WTP = (WillingToPay - mean.WTP) / sd.WTP)

# View first few rows
head(broccoli |> select(WillingToPay, z_WTP), 3)
```

```
# A tibble: 3 x 2
  WillingToPay z_WTP
         <dbl> <dbl>
1         4.34  1.36
2         3.81  1.06
3         0    -1.12
```

1.36 stand. dev. above the mean

1.12 std. dev. below the mean

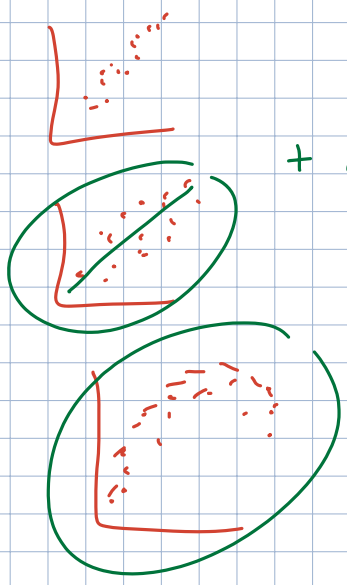
4. Features of joint and conditional distributions

Why Do We Care?

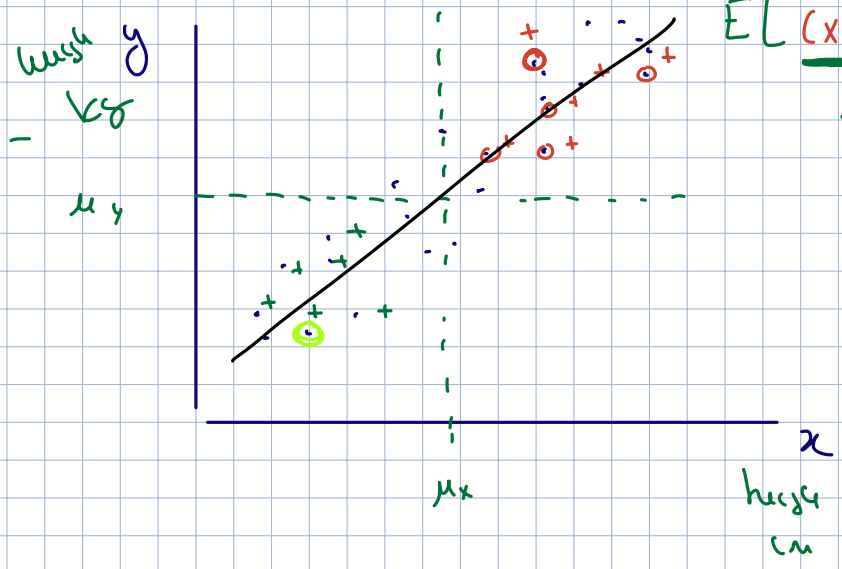
When working with **multiple random variables**, we often ask:

- How are they related?
- Does one variable help us **predict** another?
- How does the **spread** change when we condition on a known value?

This section introduces **tools to summarize** these relationships.



+ or −



$E[(X-\mu_X)(Y-\mu_Y)]$ +

+ + = +

− − = +

Covariance

The covariance measures the **direction** of a **linear association** between two random variables.

$$\sigma_{XY} \equiv Cov(X,Y) \equiv E[(X-\mu_X)(Y-\mu_Y)]$$

Covariance may range from negative infinity to positive infinity.

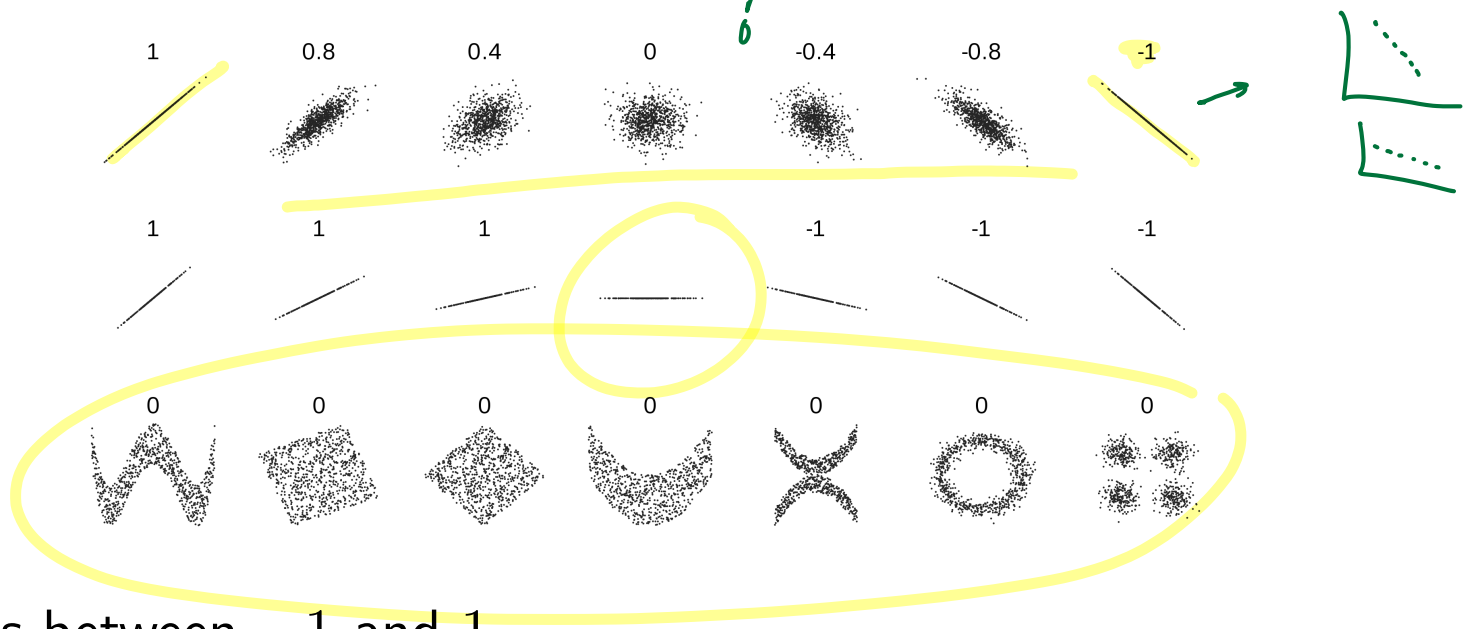
- $Cov(X,Y) > 0$ X and Y have a positive linear association.
- $Cov(X,Y) < 0$ X and Y have a negative linear association.
- $Cov(X,Y) = 0$ X and Y may be independent or have a non-linear association.

Important: If X and Y are statistically independent $\rightarrow Cov(X,Y) = 0$.
But the reverse is **not** necessarily true.

Correlation Coefficient

Covariance has a problem – its scale depends on the units of X and Y.
The **correlation coefficient** standardizes it:

$$\rho_{XY} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

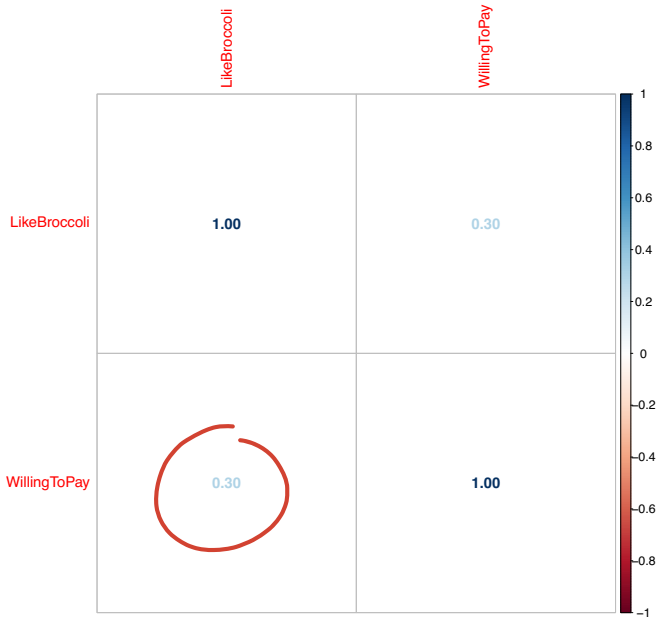


- Always between -1 and 1
- Measures **strength** and **direction** of a **linear relationship**

Correlation Matrix

```
# Load packages
library(corrplot)

# Correlation matrix
M <- cor(broccoli |> select(LikeBroccoli, WillingToPay))
corrplot(M, method="number")
```



Review:

DATA GENERATING PROCESS → Wage, $E[\text{Wage}] = \mu$

- Random Variables: X, Y, Z (x, y, z) — μ
- Probability Dist.
- Relationships
- Noise

Features of Prob Dist.

One variable (marginal dist.)
- $E[X] = \mu$
- $V(X) = \sigma^2$
  standard deviation: σ

Standardizing: $z = \frac{X - \mu}{\sigma}$

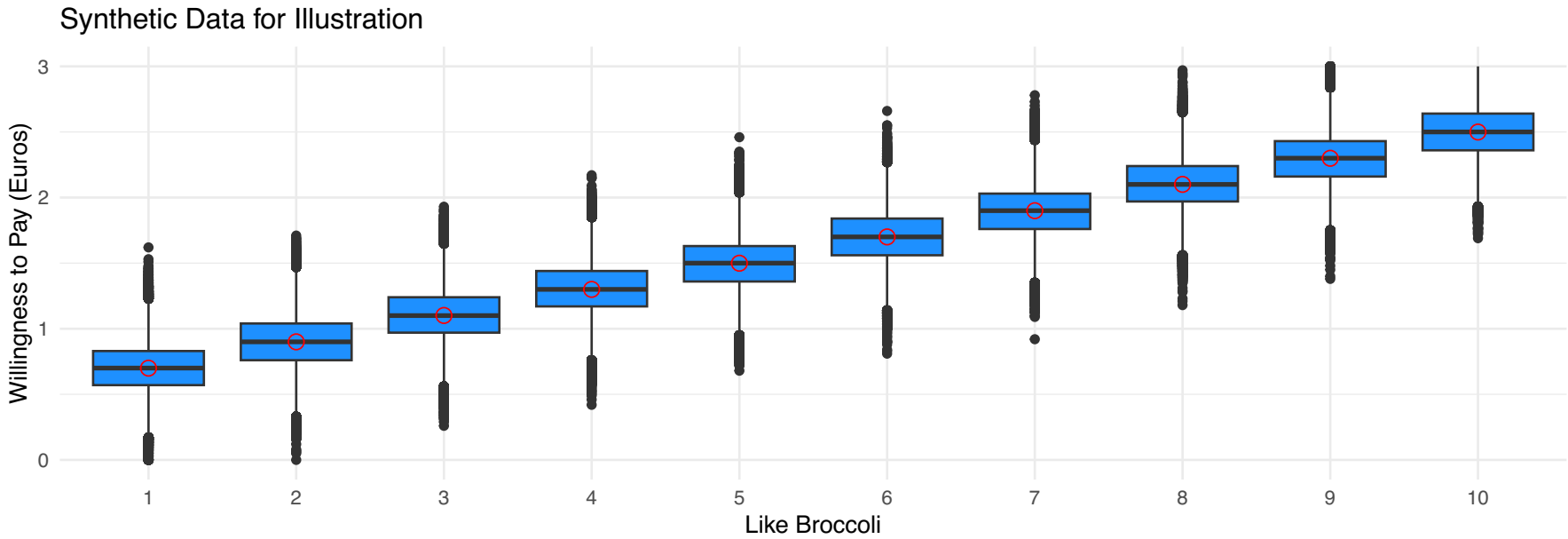
Two variables (joint dist.)
- Covariance: direction + or −
- Correlation: direction and strength
  -1 to 1

  $\rho = \frac{Cov(x, y)}{\sigma_x \times \sigma_y}$
- Conditional Mean / Expectation
- Conditional Variance

Conditional Expectation (or Conditional Mean)

Oftentimes, we want to explain a variable Y in terms of another variable. One way to summarize this information is to calculate the **conditional expectation**, the mean of Y given X:

$$E[Y|x]$$

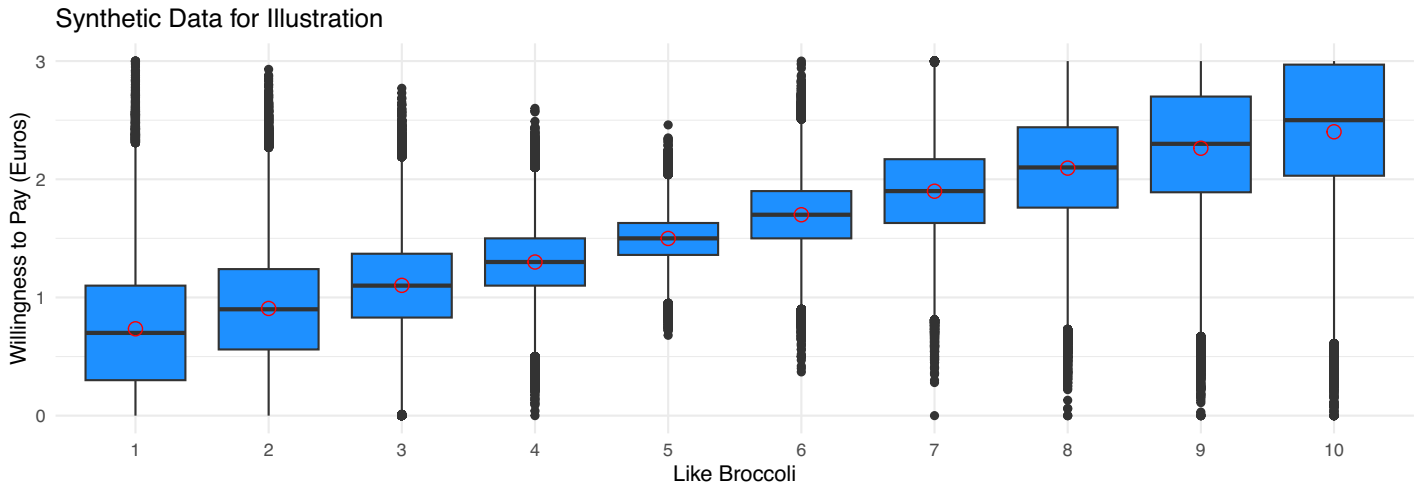


Note: If X and Y are independent, then $E[Y|X] = E[Y]$

Conditional Variance

The **conditional variance**, $Var(Y|X = x)$, measures the variability of a random variable given that another variable is fixed at a specific value.

In other words, it quantifies how much the outcome of one variable fluctuates when we know the value of another variable.



Note: If X and Y are independent, then $Var(Y|X) = Var(Y)$

5. Common probability distributions

Why Common Probability Distributions?

There are **infinitely many** possible distributions... but **real-life data** often follow **typical patterns**.

We'll study the most common ones, because they show up **again and again** in real data and in the statistical methods we use:

- A **bell-shaped curve** for heights, test scores, etc. (Normal, t-distribution)
- A series of **yes/no outcomes** like buying or not buying (Bernoulli/Binomial)

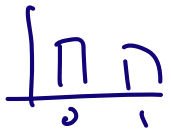
These distributions give us a simple way to:

- Understand **how data tend to behave**
- Make predictions
- Build statistical tools for hypothesis testing

Bernoulli Distribution

2 possible outcomes { Success 1 P(Success) = p ; Failure 0 P(Failure) = 1 - p

x	P(x)
0	1 - p
1	p



Let $X \sim \text{Ber}(p)$ or $X \sim \text{Bin}(1, p)$

We use the Bernoulli distribution when there are **two outcomes**:

- Success (with probability p)
- Failure (with probability $1 - p$)

We define a random variable:

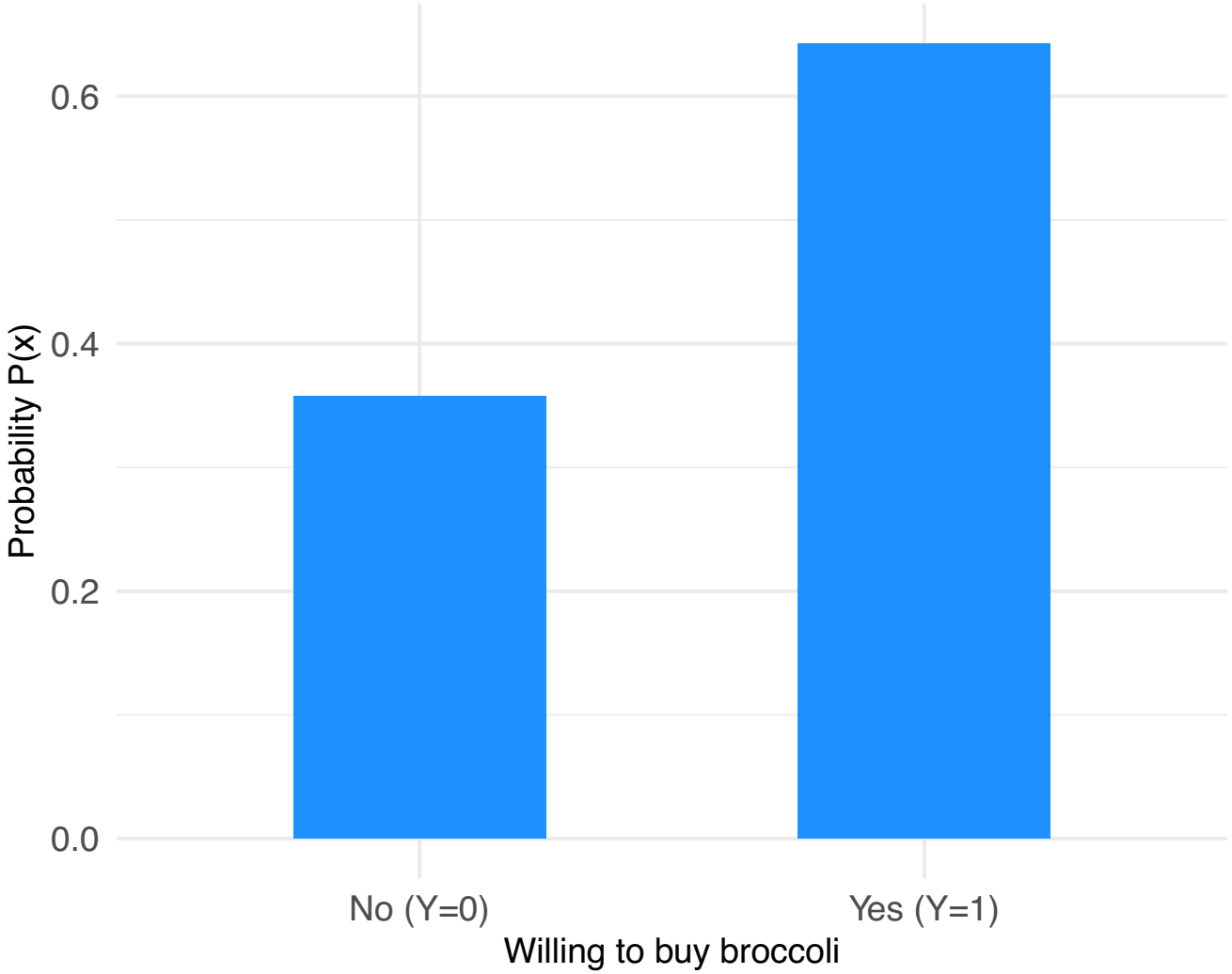
- $X = 1$ if Success
- $X = 0$ if Failure

$P(x) = p^x (1-p)^{1-x}$

$$P(X = x) = p^x (1-p)^{1-x}, \quad x \in \{0, 1\}$$

- **Mean**: $E[X] = p$ → prob. of Success → proportion of successes in the pop.
- **Variance**: $Var(X) = p(1-p)$

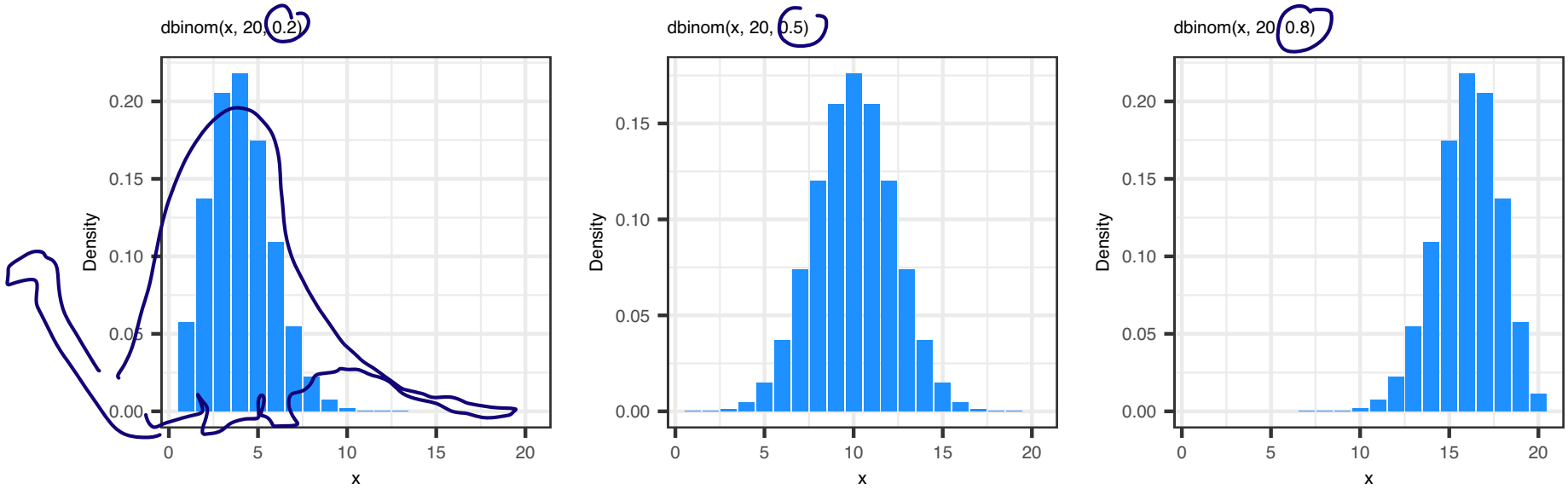
Distribution of Variable "WillingToBuy"



Binomial Distribution $X \sim Bin(n, p)$

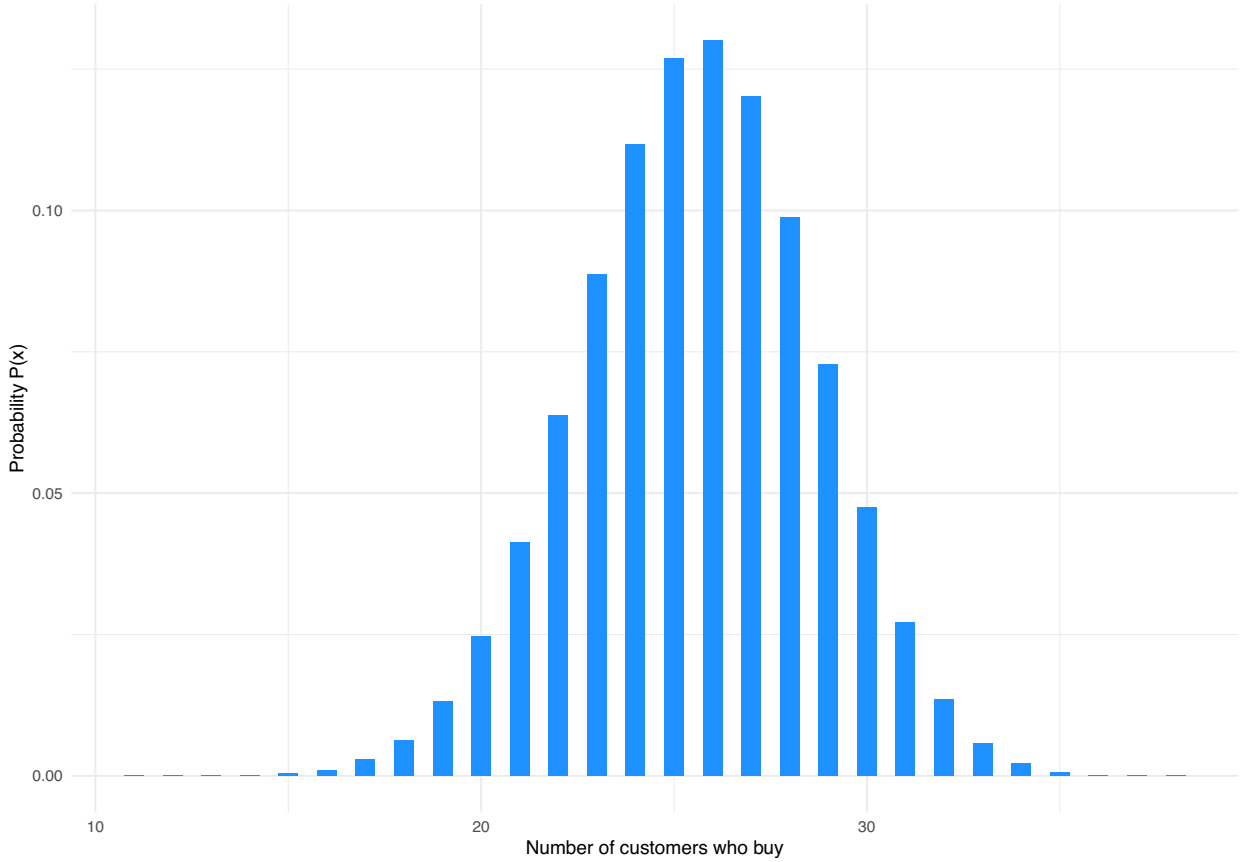
The **Binomial distribution** models the number of successes in n independent trials of a Bernoulli process.

- Its **mean** is $E[X] = np$
- Its **variance** is $Var(X) = np(1-p)$.



Distribution of Nr. of Customers Who Buy in Sample

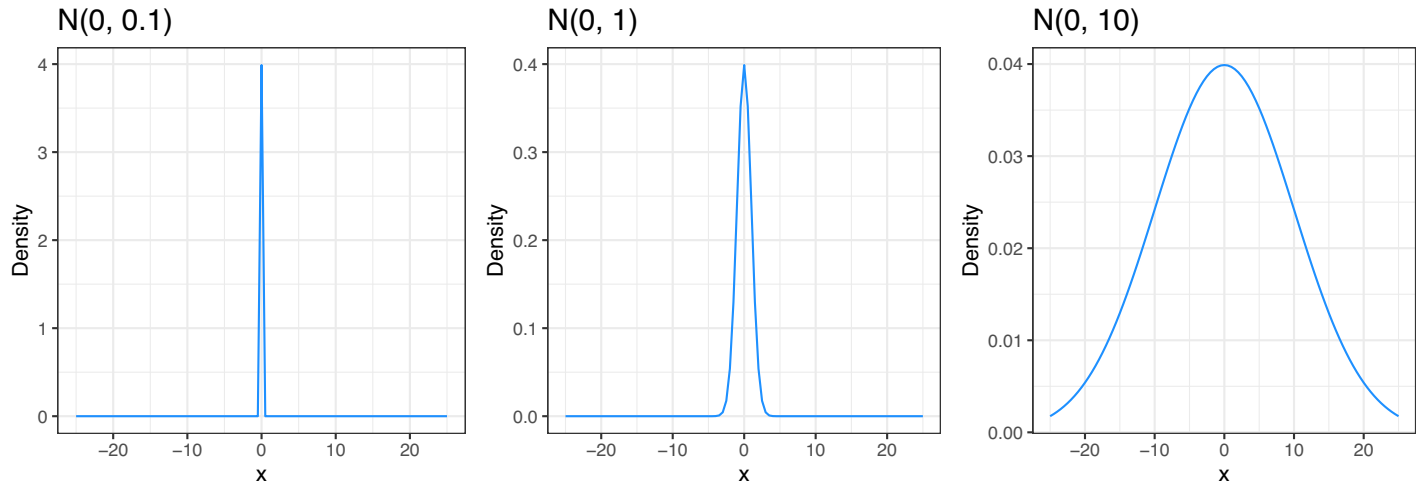
- Suppose 64% of customers are willing to buy broccoli at €1.31.
- What does the distribution look like in a random sample of 40?



Normal Distribution $X \sim N(\mu, \sigma^2)$

If X is a normally distributed random variable, $X \sim N(\mu, \sigma^2)$, then:

- Its **mean** is: $E[X] = \mu$
- Its **variance** is: $Var(X) = \sigma^2$

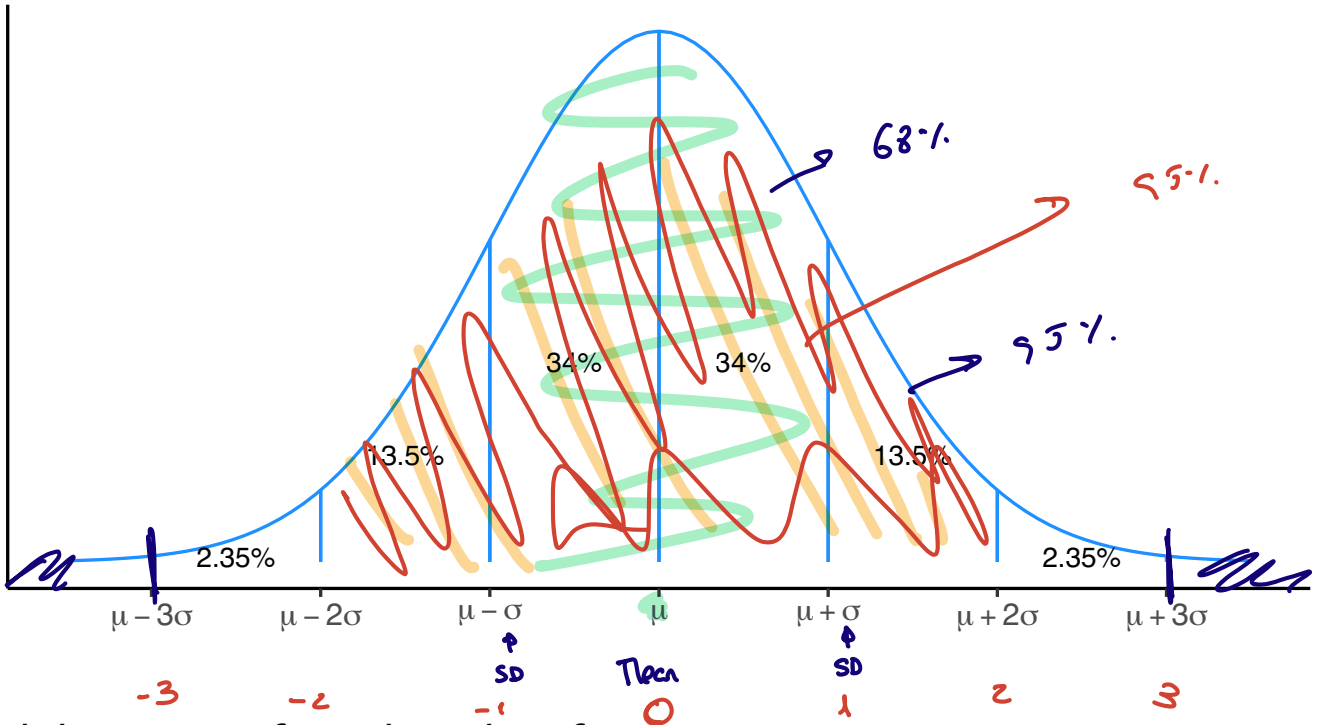


A special case of the normal distribution is the **standard normal distribution**: $Z \sim N(0, 1)$

$\mu = 0 \quad \sigma^2 = 1$

Empirical Rule

For any normal distribution with mean μ and variance σ^2:



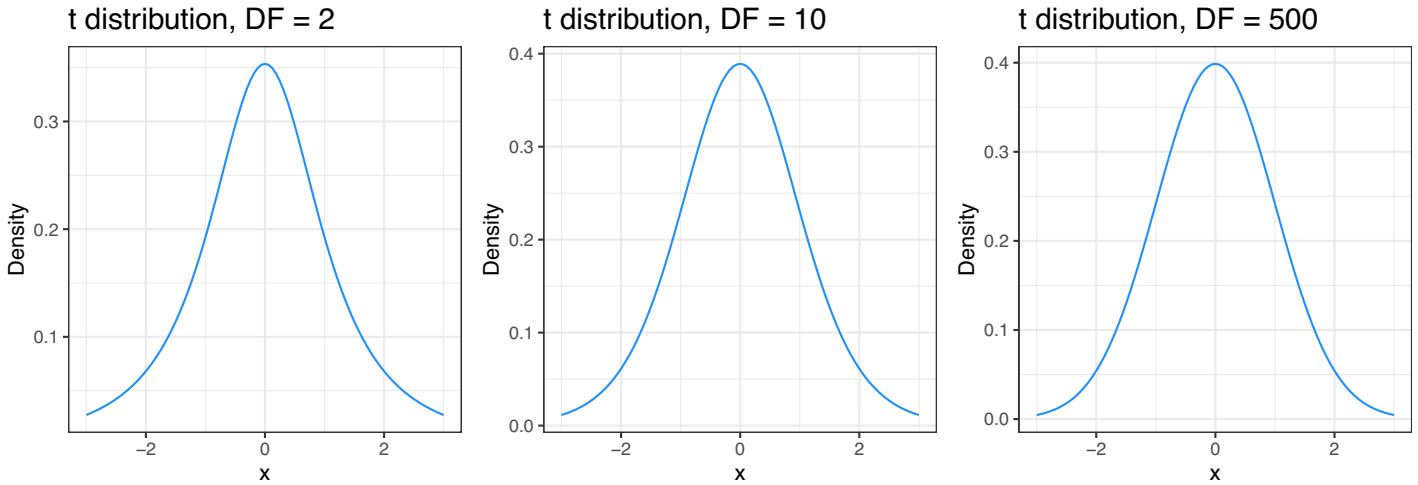
These probabilities are often thought of as:

- $P(\mu - \sigma \leq X \leq \mu + \sigma) = 68\%$
- $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = 95\%$
- $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = 99.7\%$

-1.96 , 1.96

The Student's t Distribution $X \sim t_n$

- The t distribution is the main distribution used for inference in econometrics.
- As the **degrees of freedom increase**, the t-distribution tends towards the **standard normal** distribution.
- Like the standard normal distribution, the t distribution is symmetric about zero but has heavier tails and a lower peak.



6. Wrapping Up

What Is a Probability Model?

Let's connect the dots:

- A **random variable** represents uncertainty — not just a data column
- A **probability distribution** is a model for how that variable behaves
- **Data** is just one realization from this underlying process

Tools like **regression analysis** and **statistical inference** rely on these foundations:

- To model relationships between variables
- To quantify uncertainty in estimates
- To assess whether observed patterns are meaningful or just random variation

We Studied...

- Random variables $\rightarrow$ to formalize uncertainty
- Probability distributions $\rightarrow$ to model it
- Expectations, variance, correlation $\rightarrow$ to summarize it
- Common models:
  - **Bernoulli** = binary choice
  - **Binomial** = repeated trials
  - **Normal** = foundation of inference
  - **t** = small-sample regression workhorse

These aren't just abstract ideas, they're the building blocks behind the models you'll use to analyze data and make decisions.

Recommended readings

- Wooldridge, *Introductory Econometrics*, Math Refresher B: Fundamentals of Probability
  - B-1 Random Variables and Their Probability Distributions
  - B-2 Joint Distributions, Conditional Distributions, and Independence
  - B-3 Features of Probability Distributions
  - B-4 Features of Joint and Conditional Distributions
  - B-5 The Normal and Related Distributions